Electric Propulsion Propellant In-Space Transfer Demonstration

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Applicant details

Welcome to the NASA TechLeap Prize Robotically Manipulated Payload Challenge

Before you can begin a submission, you must complete registration. The registration form identifies your lead individual or lead organization and collects the contact information NASA and the challenge administrator need to communicate with you throughout the challenge.

Registration must be completed by **5:00 p.m. ET on Wednesday, July 29, 2026**. After this deadline, new registrations will not be accepted, and you will not be able to start a submission.

Once you have completed the registration questions below, you will be able to begin your submission. To edit your registration information later, return to your profile.

Refer to the rules, terms, and conditions for a complete set of eligibility requirements. All fields are required unless otherwise noted.

Review the FAQs if you have questions, and contact hello@nasatechleap.org for technical support.

Registration type	Individual
Legal name	Liem Bui Fairbairn
Website or social media page	Not Applicable
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Size of lead organization	Less than 50 employees
How did you hear about the NASA TechLeap Prize?	✓ Other
Check here if you would like to receive open innovation updates from the challenge administrator.	✓

Submission details

Payload demonstration description

> The demonstration is led and developed by Liem Fairbairn, MEng Aeronautical Engineering undergraduate at Imperial College London. The payload uses the FFR arm to transfer xenon gas from a supply to a receiving chamber, demonstrating the actuation and transfer procedure between supercritical and vacant chambers.

Capabilities

> Proficiency in programming (MATLAB and Python) will allow for the development of the program governing electronic processes. Further experience with small scale manufacturing can be evidenced with the group design and manufacturing of a 2 blade wind turbine (June 2026), in which I worked dominantly on aerodynamics and manufacturing. Access to the university workshop and equipment yields the ability to assemble the demonstration, and university access to FEA and CFD software allows for the further amendment of the demonstration via computational analysis if necessary. Moreover, proximity to professors and senior students allows for advisory measures to be taken during the final design and manufacturing process.
>
> My most relevant experience is the 'Mission discovery' design engineering competition for operating mechanisms in microgravity conditions. I designed a simple electromagnet tensile tester which would assess changes to tensile behavior of spider silk produced in microgravity, with hopes of investigating sustainable material production in extraterrestrial habitats, with medicinal application. While not a winning project, it has exposed me to the space engineering environment, where I have been keen to be involved since.
>
> Other past experiences relevant to this demonstration include:

Experimental and Computational pipe flow analysis to validate theoretically derived models. Structural analysis of simple structure in Abaqus, comparing against theory.

Urban Electric Vehicle Design and CAD development

Aforementioned design and manufacturing of a wind turbine, wind tunnel tested at 1400 rpm

Team size	Less than 10
Proof of citizenship, permanent residency, or primary place of business	✓ I can provide proof of citizenship or permanent residency.
Proof of liability insurance	Yes, I confirm I have at least $250,000 in liability insurance coverage or can demonstrate equivalent financial responsibility.

Payload demonstration overview

In alignment with shortfall 23.07, the demonstration considers the transfer procedure of xenon gas from a supercritical supply tank to a mock receiving fuel tank, where the transfer is controlled by a pressure gradient, and the autonomous control of a ball valve. Using the passive crosslink cleat, the supply tank is moved by the FFR arm, to attach to a receiving tank, connecting by means of an electromagnetic locking mechanism to secure the mating components. A male dogbone component slots into a female rectangular gap, whereby a switch is hit at the base of the penetration point, engaging magnetic rectangular locks to close around the male component. Upon the connection, a microcontroller is alerted of the attachment, and initiates a motor which opens the ball valve, allowing for xenon gas to pass between chambers.

The possibility of an off the shelf nature for both the motor and valve yields only the attachment mechanism and chambers to be manufactured, within the capabilities of myself.

Within the demonstration, two pressure transducers and temperature sensors are placed within each chamber, feeding into an arduino. Information is then fed through a MATLAB script which will output temperature and pressure curves as functions of time, then plotted against the modelled situation, providing necessary information required to amend the model when scaled up for true size spacecraft. An equation of state script then can interpret the mass flow rate, another metric which can be measured for transfer times of propellants. Aiming for around 10-20% tolerance given the nature of the demonstration seems appropriate, and provides a measurable success criterion.

Mass accommodation has prevented the addition of a second valve, so the demonstration concludes by a rerelease of propellant back into the supply, whereafter the valve can be resealed and the demonstration is complete.

Payload demonstration benefits

With recent NASA investigation into nuclear powered ionic propulsion (IP), there is clear motivation to develop mechanisms to transfer ionic propellant in microgravity conditions. Beyond the standalone benefits of IP, that being extreme fuel efficiency, long lifetime, and thus a better mass economy, the transfer of IP propellant would allow for the reinvigoration of deep space missions, providing spacecraft perhaps dormant in distant orbit, with the necessary fuel to continue exploration, a concept in alignment with shortfall 23.07. Moreover, the propellant independence in this demonstration yields progress in shortfall 23.03 inevitable. Dedicated propellant transportation aircraft would make use of the proposed demonstration as a proof of concept to the supercritical transfer of Xenon, without the need for larger, complex cooling systems that might otherwise be found in cryogenic storage methods. Alternatively, this would permit the full fueling of IP powered spacecraft in orbit, be that in extraterrestrial habitat, or in orbit. Extraterrestrially stored xenon may then be transferred to aircraft which may enter orbit in number, before being fueled, thus providing efficient martian transportation, or beyond.

Future amendments to payloads similar to this may further yield progress in 23.05 and 23.06, be that through material/structural changes or through staggered transfer. Irrespective of even this, the proposed payload is largely elastic in its application and puts forward a foundational template for successive investigation

An in-orbit robotic arm carrying out this task increases the tolerance of refueling positions, allowing for more corrective positioning for robotic refuelling, also accommodating for a range of spacecraft where refueling may change due to fuel tank positioning. This flexibility allows for more variability in spacecraft design in the future.

Moreover, the data set provided is largely novel, that being flight qualified data regarding microgravity transfer of supercritical xenon. The contemporary nature of refueling logistics would benefit from a 2027 flight test of this demonstration, as decisions have potential to be more informed.

Visual representation

PDF

Visual Rep Submission.pdf (938 KiB download)

Payload development status

Much of the storage concept development is derivative of Richard P. Welle's 1991 paper Propellant Storage Considerations for Electric Propulsion, from which the standards of optimal mass density and pressure were developed and integrated into this demonstration.

> The novel locking mechanism can be tested and modified well within the 8 month production window - modifications to exact dimensions can be easily machined.
>
> I developed an equation of state matlab model, in accordance with Welle's mass optimization work to test general thermodynamic properties of the flow, such as temperature and pressure changes during transfer and storage, informing the tank thickness and material selection. Uncertainty in the model forced me to make a worst case decision with a safety factor of 3, regarding receiving tank thickness, sacrificing some mass reduction. This can be amended and further investigated during manufacturing, and re-evaluation of mass balances.
>
> The CAD model has been briefly developed in Dassault 3DX for a general sense check with design dimensions.
>
> For the model developed, safety factor of 3x was chosen as a practical middle ground given manufacturing constraints, between the NASA-STD-5001proof (1.5×) and analysis-only burst (4×) requirements

Benefit of hosted orbital flight

> This provides a strong reference point for the thermodynamic implications and design challenges of operating without convection and conduction and ways of managing heat transfer. The dominant design challenge proved to be implementing a heat transfer model into the iterative solving of mass transfer differential equations, as it was otherwise a source of nonphysical values.
>
> Moreover, fluid behaviour in microgravity is an area of challenge in replication on Earth, thus conclusive concrete data on the behaviour of gas in space conditions is, of course, best found in orbit. This is mainly found with the gravitational effect on convection, where temperature and density variances cause currents which disappear in space conditions. Thus the payload's data will be thermodynamically unique from earth bound testing.

Payload development plan

> The use of passive crosslink cleat 3 demands a surface of maximum 350mm by 350mm as shown in the attachment to the supply tank concept in the provided images and video proposal. The machining of the aluminium is likely to be achieved through access to university machinery, where spherical components can be molded in halves via casting. The process can be individually carried out, as the mechanical complexity has deliberately been kept to a minimum to facilitate this. Mass optimization has kept components down to below (approximately) the 12.5kg limit, with a total tank mass of 1.46kg (approx), and rough mass of 5kg for the valve transfer mechanism. The additional weights of the electronics governing the locking mechanism is generally lower, however the weight of the locking mechanisms remains in speculation, due to the connection strength and tolerance being undecided as the ability to physically test this is not yet decided. In the event that the proposed system is unworthy, an external mode of connection may be employed by off the shelf purchase.

Points of risk arise in the manufacturing of the magnetic connection system, as manufacturing this system proves to be the largest design challenge and thus projected to be a large manufacturing challenge. General safety concerns are standard with solo working, namely traditional workshop hazards, or proficiency using machinery, all of which I may receive training in, in the case proficiency in the used machinery is critical to the success of the demonstration production.

Intellectual property

All designed systems of connection, as well as tank design drawings are developed and owned by myself - Liem Fairbairn. Off the shelf use remains in partial speculation, and thus is inconclusive, however the use of a ball valve and motor poses no significant legal difficulties and thus has not been deemed a sufficient point of interest to consider at this stage.

Imperial College holds responsibility for machinery used and any further advice received will be duly credited where appropriate.

Other considerations

It is worth noting that the thermodynamic uncertainty due to storing xenon at supercritical conditions meant that my MATLAB model failed to produce accurate results, rather providing a lower and upper bound of maximum pressure in the receiving tank, and thus, using a safety factor of 3, the thickness of the receiving tank was chosen to be 11.8mm. The flow being initially choked due to high pressure differences requires lower transfer rate to mitigate vibration. However flow model uncertainty forces an upper bound to be used for receiving tank thickness.This was because the receiving tank was made not to optimise mass, but to be able to sustain maximum expected pressure during the transfer process, due to adiabatic heating effects.

Solo working is a general constraint, as this drastically increases production time and induction will be necessary for a number of manufacturing processes, further potentially delaying the project.

YouTube video URL

https://youtu.be/5oMebA2QMLI

Project plan

Consult the workshop technician early regarding machinery access, booking slots, and any safety training required for casting and machining aluminium.

Place off-the-shelf orders immediately i.e. ball valve, gearbox, stepper motor, locking magnets, arduino, pressure transducers, and temperature sensors.

Purchase aluminium stock and prepare moulds for the tank halves. This is expected to take a few iterations given limited prior moulding experience, and is flagged as the highest-risk item on the timeline, so most schedule margin sits here - likely a month at most.

Once cast, machine the tank halves and the locking mechanism body, embedding the electromagnets and switch system at this stage.

Organise the electronics governing the locking mechanism and write the arduino/microcontroller program; this is expected to take a month to a month and a half, as the final electronics configuration isn't yet locked in.

Add and position the pressure/temperature sensors within each chamber.

Once assembled, proof-test the tanks with high-pressure water above the 8.3 MPa expected pressure, then move to gas leak testing across the full assembly if that passes.

Full process expected to take 5–6 months, leaving 2–3 months of margin within the 8-month window for remodelling or design changes if necessary.

Total cost	US$4,700.00

Budget

Costs were estimated from current commercial pricing for aerospace-grade aluminium stock, off-the-shelf actuator/valve components, and current xenon market rates, organized by phase

Phase 1 — Preparation and Casting (Month 1)

Aluminium billet/bar stock (tank halves + locking mechanism, incl. machining waste): £20–40

Xenon gas, research grade, ~0.74 kg + margin: £800–2,000

Off-the-shelf ball valve + gearbox + stepper motor: £150–400

> Locking magnets/electromagnets + switch: £30–80
>
> Arduino, pressure transducers (×2), temperature sensors (×2), wiring/connectors: £150–250
>
> Fasteners, seals, high-pressure fittings, quick-disconnect components: £100–250
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> Casting consumables (mould material, release agent): £50–150
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> Phase 2 — Fabrication & machining (Months 1–2)
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> Machine shop consumables (cutting tools, tooling wear): £50–150
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> Machinery safety training/certification, if required: £0–100
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> Contingency for casting iterations (extra stock for scrapped attempts): £30–60
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> Phase 3 — Electronics integration (Months 2–3.5)
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> PCB/protoboard, wiring harness, connectors: £30–80
>
> Sensor calibration consumables: £20–50
>
> Phase 4 — Assembly, proof test & qualification (Months 3.5–6)
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> Pressure test consumables (seals/gaskets replaced post-test): £30–80
>
> Leak-check consumables: £20–50
>
> Overall Budget - 3300 - 3500 GBP

Budget narrative

> N/A

I have not received government funding for the aspects of the proposed payload I aim to develop as part of this challenge.	✓
I have read and understand the official rules, terms, and conditions, and fully agree to them.	✓
I am an individual, represent a team of individuals, or represent a legal entity that meets all eligibility criteria as stated in the official rules, terms and conditions.	✓

I did not receive assistance from challenge sponsors, judges, or any other individuals associated with the challenge in preparing this submission.	✓

Log in to rmpc.awardsplatform.com to see complete submission attachments.